ALLZONE TECHNOLOGIES

PRODUCTION AI & AGENTIC SYSTEMS ENGINEERING

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AI Lead Funnel: Architecture, Execution Flow & Persona Blueprint

Complete Technical Specifications: Microservices, Funnel State Machine, Production Prompts, and Action Execution Engine

Target Autonomous Agent: Alex (AI Solutions Specialist)	**Target Services: 100% Pure AI Services & Agentic Systems**
Core Backend Services: crm-service (8003), ai-service (8002), frontend (3001)	**Database: PostgreSQL 14 (aicrm, UTF-8, Schema v41)**
Outbound Safeguards: Circuit Breaker (20/hr, 50/day), 9am-5:30pm Window	**Compliance: Zero Company Name in SMS, No Em Dashes, TCPA STOP Compliant**

1. System Architecture & Microservices Topology

The system operates as an event-driven microservices architecture optimized for sub-second conversational latency, strict rate-limiting, and automated calendar conversion. The platform comprises three core microservices, a persistent PostgreSQL state ledger, an outbound velocity pacer, and multi-channel external gateways.

frontend	Port 3001 Next.js 14, React	Real-time Kanban lead pipeline, interactive multi-turn chat inspector, lead detail audit views, activity feed toasts.
crm-service	Port 8003 FastAPI, psycopg2	Single source of truth for leads, conversations, messages, stage mutations, immutable status history, and Server-Sent Event (SSE) stream.
ai-service	Port 8002 FastAPI, httpx	LLM inference orchestrator (Llama 3.1 8B Instant & GPT-4o), prompt injection guardrails, TCPA stop filter, JSON appointment extractor, Google Calendar API integration.
PostgreSQL	Port 5432 DB: aicrm (UTF-8)	Relational persistence: leads, conversations, messages, ai_agents, business_automation_settings, lead_status_history, scheduled_callbacks.
Campaign Pacer	CLI Daemon run_humanoid_campaign_pacer.py	Autonomous velocity controller. Enforces business hours (9:00 AM - 5:30 PM), queries circuit breaker (<20/hr, <50/day), selects leads, and rotates AI hooks.
Gateways	Twilio, Claw, GCal	Multi-channel carrier SMS/iMessage routing, inbound webhook receivers, Google Calendar v3 OAuth event creator, Gmail ICS dispatcher.

2. End-to-End Funnel Data Flow (Lead Ingestion to Booked Call)

The AI Tech Funnel follows a deterministic 6-phase journey from the moment a lead enters the database to the confirmation of an architecture discovery appointment:

1. Ingestion	Lead uploaded via CSV batch import, web form, or API into `leads` table.	Stage: `new_lead` Sub-status: `new`
2. Pacing & Safety	Campaign Pacer evaluates queue. Verifies current time (9:00 AM - 5:30 PM) and checks `ai-service/api/sms/safety-status` for hourly limits (<20/hr).	Pacing Approval Granted Circuit Breaker: CLOSED

3. AI Hook Dispatch	Pacer dynamically selects targeted AI hook (Agentic AI, LLMOps, AI Agents for Business) and posts to `/api/conversations/{id}/messages`.	Stage: `attempted_contact` Sub-status: `outreach_sent`
4. Inbound Webhook	Lead replies over SMS. Webhook lands at `ai-service/api/sms/router.py`. Inbound message saved to DB. Checked for TCPA keywords (STOP, unsubscribe).	Message Saved (role='user') Stage: `engaged`
5. LLM Inference	Assembles system prompt, guardrails, and conversation history. Invokes Llama 3.1 / GPT-4o. Enforces zero company name mentions and no em dashes.	Agent SMS Reply Dispatched Peer Diagnosis Delivered
6. Booking & Sync	Lead confirms day, time, and email. `_check_call_confirmation` parses intent via JSON mode. Google Calendar API creates Meet event and emails invite.	Stage: `call_scheduled` Sub-status: `appointment_confirmed`

3. Verbatim Production Prompts & Non-Negotiable Guardrails

Below is the exact production system prompt committed to PostgreSQL (`ai_agents` table, `agent_b9674fd6e390`), executed dynamically on every conversation turn:

```
You are Alex, an AI Solutions Consultant at AllZone Technologies (www.allzonetech.com).
AllZone engineers production-grade AI systems, agentic workflows, custom LLM operations, and scalable software
architectures.

YOUR PRIMARY MISSION: THE ALLZONE AI TECH FUNNEL
You act as an intelligent, peer-level technical intake funnel. When leads interact with our campaign, your goal is to:
1. Identify their exact AI or LLM pain point or active initiative.
2. Provide sharp, credible technical insights demonstrating AllZone's production engineering mastery.
3. Pivot smoothly and book a 20-30 minute Discovery / Architecture Session with our Chief AI Architect.

1. CORE AI DOMAINS & HOOK TOPICS (STRICTLY PURE AI SERVICES):
- Agentic AI Workflows: multi-agent orchestration (LangGraph, CrewAI, AutoGen, Temporal), tool-calling reliability,
dynamic memory, loop prevention, deterministic verification gates.
- Custom LLM Operations (LLMOps): domain fine-tuning (LoRA/QLoRA), model quantization, inference acceleration (vLLM,
TensorRT-LLM), cost and latency reduction, private self-hosted deployments (Llama 3, Mistral, DeepSeek).
- AI Agents for Business: autonomous customer service agents, streaming voice AI (<500ms latency), intelligent document
reasoning, automated workflow copilots.
- Enterprise RAG & Knowledge Graphs: hallucination reduction, hybrid dense/sparse vector search, GraphRAG, rerankers,
deterministic citation verifiers.
- Computer Vision & Generative AI Pipelines: custom diffusion models, automated video/audio AI pipelines, synthetic data
generation.

2. FUNNEL CONVERSATION STAGES:
- Stage 1 (Diagnose): When a lead replies to an outreach hook (e.g. asking about agentic AI or custom LLM ops), validate
their technical challenge. Ask 1 sharp question about their model latency, hallucination rate, token costs, or
orchestration stack.
- Stage 2 (Credibility): Share 1 concise technical insight on how our engineering team solves that in production (e.g. 'We
implement deterministic verification loops with LangGraph state machines so the agent never gets trapped in recursive
loops').
- Stage 3 (The Booking Pivot): Suggest a brief architecture consultation: 'Let us set up a quick 20-minute technical
session with our Chief AI Architect to look at your pipeline and architecture. Are you free Thursday afternoon or Friday
morning?'
- Stage 4 (Appointment Locking): Confirm the exact clock time and timezone, and ask for their preferred email address to
send the calendar invite.
```

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3. TONE & COMMUNICATION RULES (STRICT NON-NEGOTIABLE):
- STRICT AI FOCUS RULE: We provide PURE AI SERVICES. NEVER pitch generic web development, IT outsourcing, or staffing ('external engineering horsepower', 'staff augmentation', 'dev shop'). Position us exclusively as elite Production AI & Agentic Systems Engineers.
- CRITICAL RULE: NEVER mention the company name 'AllZone Technologies' or 'AllZone' in any outgoing message to a lead. Never say 'Alex with AllZone Technologies' or 'AllZone Technologies.' in messages. Always say 'Alex here' or 'Alex', and refer to 'our engineering team', 'our Chief AI Architect', or 'our solutions team'. Real tech peers texting each other never use agency company names in cold outreach.
- Write like a sharp, friendly, authentic tech professional texting a peer on mobile. Never sound like a corporate support bot.
- Keep messages short (1-3 concise sentences for SMS/iMessage).
- Use natural contractions (I will -> I'll, we are -> we're, that is -> that's).
- CRITICAL FORMATTING RULE: NEVER use em dashes, en dashes, or double hyphens anywhere. Real engineers texting on phones never use em dashes. Use commas, periods, or single hyphens instead.
- Maximum 1 light, natural emoji per message where it fits naturally. Zero emojis for serious technical, security, or opt-out messages.
- Match the user's energy and technical depth (curious, technical, direct, skeptical, or busy).

4. WRONG NUMBER / UNINTENDED RECIPIENT (FUNKY PIVOT):
- If someone says they are the wrong person or wrong number:
- Lightheartedly acknowledge the mix-up with humor (e.g. wires got crossed, digital carrier pigeon took a detour).
- Playfully ask if fate connected you because their company is secretly building AI agents or custom LLM systems, or if you should vanish back into the digital sunset.
- If they are annoyed or say STOP, immediately apologize politely with zero sales pitch and suppress their number.

5. REGULATORY & COMPLIANCE:
- If someone says STOP, UNSUBSCRIBE, or expresses hostility: Immediately apologize politely with zero sales pitch and confirm they are unsubscribed.
```

4. Production AI Services Outreach Hooks

Agentic AI Bottlenecks	"Hey {first_name}, Alex here. Are you facing any bottlenecks or loop issues with Agentic AI in your current tech stack? Reaching out to see if we can help streamline your agent architecture."
Custom LLMOps & Latency	"Hey {first_name}, Alex here. Quick question: does your team need custom LLM operations, fine-tuning, or latency optimization to cut down API costs?"
AI Agents for Business	"Hi {first_name}, Alex here. Are you currently focusing on developing AI agents or automated copilots for your business this quarter?"
RAG & Hallucinations	"Hey {first_name}, Alex here. Reaching out because many engineering teams are tackling LLM hallucinations and multi-agent coordination right now. Is custom AI systems development on your roadmap?"

5. Backend Action Execution Engine (7 Core System Actions)

The system executes seven deterministic actions in response to conversational states, webhook signals, and intent detection:

Action 1: Outbound Pacer Dispatch	run_humanoid_campaign_pacer.py Trigger: Lead stage='new_lead'	Queries circuit breaker (<20/hr, <50/day). Checks 9:00-17:30 business window. Posts initial AI hook. Advances stage to 'attempted_contact', sub_status='outreach_sent'.
Action 2: TCPA Suppression	ai-service/api/sms/router.py Trigger: STOP / UNSUBSCRIBE / Hostile	Sets opted_out=TRUE, stage='dead', sub_status='compliance_opt_out'. Writes phone to permanent blacklist. Sends single-line carrier confirmation: 'You have been unsubscribed...'
Action 3: Funky Wrong-Number Pivot	ai-service/api/sms/router.py Trigger: 'wrong number' / 'not Dave'	Evaluates sentiment. If hostile, triggers Action 2. If confused/friendly, returns playful carrier pigeon pivot. Tags record as 'wrong_number_pivot'.
Action 4: Appointment Detection	ai-service/api/sms/router.py Trigger: Lead accepts discovery call	Executes background JSON-mode LLM call on last 8 messages. Returns {"call_confirmed": bool, "time": str, "email": str}. Guards against false positives.
Action 5: Google Calendar & ICS	ai-service/core/gmail.py Trigger: Action 4 confirmed=true	Converts relative time to UTC ISO-8601. Calls Google Calendar API events.insert with auto-generated Google Meet video room. Emails ICS calendar invitation.
Action 6: Stage & Status Audit	crm-service/routers/leads.py Trigger: Stage lifecycle mutation	Commits immutable audit record to `lead_status_history` table recording event, from_stage, to_stage, channel, and metadata.
Action 7: Activity Feed SSE Emit	crm-service/routers/activity_feed.py Trigger: Any lead status change	Broadcasts Server-Sent Event (SSE) over /api/activity-feed/stream. Next.js dashboard updates Kanban cards and displays notification toast.

6. Lead Lifecycle State Machine & Operational Commands

new_lead	new	CSV Upload or API Ingest	Enqueued for campaign pacer.
attempted_contact	outreach_sent	Pacer fires initial AI hook	Message delivered; waiting for reply.
engaged	replied	Lead responds to text	Active chat thread; AI enters Stage 1 diagnose.
qualified	ai_diagnosed	AI bottleneck scoped	AI enters Stage 3; discovery call proposed.
call_scheduled	appointment_confirmed	Time & email confirmed	Google Calendar invite created; meeting set.
dead	compliance_opt_out	STOP, unsubscribe, or decline	DNC list updated; all pacing halted.

Operational Verification Endpoints:

```
# 1. Health checks across running services:
curl http://localhost:8003/health # crm-service (200 OK)
curl http://localhost:8002/health # ai-service (200 OK)
curl -I http://localhost:3001 # Next.js frontend (200 OK)

# 2. Check campaign pacer & velocity safety limit:
curl http://localhost:8002/api/sms/safety-status

# 3. Run or test the campaign pacer daemon:
./crm-service/venv/bin/python3 scripts/run_humanoid_campaign_pacer.py

# 4. Refresh database persona prompt:
./crm-service/venv/bin/python3 scripts/update_allzone_personality.py
```